Course: Introduction to Information Security Analytics
Unit 1: Introduction to Information Security
The CIA Triad defines Confidentiality, Integrity, and Availability as the core pillars of sec
Information security policies establish baseline organizational rules for protection.

Unit 2: Cryptography Basics

Cryptography Basics transforms plaintext into ciphertext using mathematical keys. Kerckhoffs’s principle dictates that security must depend on key secrecy, not obscurity.

Unit 3: Symmetric Encryption & Cryptographic Hash Functions

Symmetric Encryption uses a shared private key for fast block-cipher encryption.

Cryptographic Hash Functions provide deterministic one-way data integrity verification.

Unit 4: User Authentication

User Authentication verifies digital identities using salted password hashing. Multi-Factor Authentication enforces layered verification tokens.

Unit 5: Access Control & Security Governance

Access Control regulates operations permitted to authenticated identities.

Role-Based Access Control assigns permissions to organizational roles based on the pr